

Pratyush Sudhakar

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EDUCATION

Cornell University

Bachelor of Science in Computer Science (*Minors: Mathematics & Psychology*)

Ithaca, NY

December 2026

LEADERSHIP EXPERIENCE

Head of Engineering & President, *Perfect Match (Student-led Venture)*

January 2023 - May 2025

Tech: Next.js, TypeScript, Tailwind CSS, MongoDB, Redis, AWS, CI/CD

Ithaca, NY

- Directed 4 cross-functional teams through launch, recruitment, and business planning while architecting the core matching algorithm (weighted compatibility scoring, Redis-cached ranking), scaling the platform to 5,000+ users.
- Built analytics platform delivering real-time survey insights to 5,000+ Cornellians via cached Map-Reduce aggregations, reducing latency by 93%; automated personalized emails driving 600+ new active users.

Technical Product Manager, *Cornell Digital Tech & Innovation*

January 2023 - December 2023

Tech: React.js, DynamoDB, Cypress, Kubernetes, Docker, Flutter, Agile

Ithaca, NY

- Spearheaded agile development of a cross-platform ride scheduling app with 7 developers, driving key architecture decisions on state management and offline caching.
- Architected a TDD pipeline (Cypress, Mocha, Supertest) reaching 94% code coverage; orchestrated CI/CD with GitHub Actions and Kubernetes to cut deploy time.

TECHNICAL EXPERIENCE

Full-Stack Developer, *DECA Lab, Cornell University*

January 2025 - Present

Tech: React Native, Expo, Next.js, FastAPI, SQLAlchemy, PostgreSQL, TypeScript

Ithaca, NY

- Built two full-stack audit platforms with typed REST APIs for multi-role workflows (access control, drafts, submission, reporting) with i18n and accessibility support for field workers.
- Engineered offline-first mobile apps (Legend State/MMKV, background sync) enabling auditors to start, resume, and submit field audits reliably in low-connectivity environments across localized, accessible interfaces.
- Designed normalized scoring pipelines with client-side CSV/XLSX/PDF export; hardened draft-save flows against duplicate-key failures and extended data integrity guarantees across multi-round field testing.

Computer Systems Developer, *Rizvi Lab, Cornell University*

July 2024 - September 2024

Tech: Python, Linux/Bash, IoT, Embedded Systems, Git

Ithaca, NY

- Built a PyQt touchscreen interface for BeagleBone systems enabling real-time monitoring and remote control of food processing equipment; automated operations with shell scripts and cron jobs, reducing manual intervention by 60%.

Software Engineering Intern, *rapStudy*

June 2023 - August 2023

Tech: React.js, TypeScript, Redux, Firebase, Material UI, REST APIs

Los Angeles, CA

- Built a secure song-sharing system (React, 10+ custom Firestore rules) and a Redux-powered global audio-playback component synced with lyrics display.

Data Engineering Intern, *Cornell College of Engineering*

June 2023 - September 2023

Tech: Python, Pandas/NumPy, Selenium, Next.js, TypeScript, ETL Pipelines

Ithaca, NY

- Engineered a scalable ETL pipeline (Selenium, NLP, multithreading) processing financial data for 3,000+ companies (-83% processing time), and a Next.js/SSE live dashboard (-66% latency).

VENTURES & PERSONAL PROJECTS

Halo - Wearable AI | *Swift (iOS), Python, OpenAI (LLMs), Redis, Celery, IoT*

June 2025 - July 2025

Built a real-time face-recognition app (iOS+ESP32) with Apple Vision + OpenAI; refactored code (-40%), trimmed app size (-30%), and cut backend latency from 3 min to 50 s via a Celery/Redis redesign.

AI Tool (Millennium Sponsored) | *Flask, Azure, RAG (Vector DB), LangChain*

March 2024 - May 2024

Devised a course syllabus assistant with Azure-hosted REST API, LangChain, and Zilliz cluster for vector similarity search, achieving 50ms average response time for context-aware student queries.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++, SQL, GraphQL, OCaml, Bash

Frameworks & Libraries: React.js, Next.js, React Native, Expo, Node.js, Express.js, FastAPI, Flask

Data, Cloud & Tools: PostgreSQL, SQLAlchemy, Redis, Docker, Kubernetes, AWS, GCP, Azure, Git, Terraform